

EAN-Enhancement

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1	Overview	1	3.4	Region of Interest (ROI) processing	10
1.1	Additional Support Documentation	1	3.5	False Color Stage	11
1.2	Sightline Software Requirements	1	3.6	Multi-Camera Image Blend Stage	11
1.3	Software Feature Version Requirements.....	1	4	Application Notes.....	11
2	Rendering Video.....	2	4.1	Warp Stage.....	11
2.1	Overview	2	4.2	Mitigating Scintillation Effects	11
2.2	Denoise	3	4.2.1	Scintillation Reducing Parameters	12
2.2.1	Motion Mask.....	3	4.3	Saving Enhancements	13
2.3	Custom Convolution.....	4	4.4	Replicating Effects of IR Cameras	14
2.3.1	Custom Convolution Examples	5	5	False Color.....	14
2.3.2	Custom Convolution Performance Expectations ...	5	5.1	False Color Guidelines.....	14
2.4	Contrast Enhancement	5	5.1.1	Frequently Used Modes.....	15
2.4.1	CLAHE	6	5.2	Related Sightline Commands	15
2.4.2	Local Area Processing.....	7	5.3	False Color Palettes.....	16
2.4.3	Histogram Equalization	7	5.3.1	Custom Color User Palette.....	17
2.4.4	Feature Based Histogram Equalization	8	5.3.2	Custom User Palette with Pre-existing LUT	18
2.4.5	Histogram Equalization + Gamma correction	8	5.3.3	Multiple Color Range Example.....	19
3	Square Root Histogram Equalization	8	5.3.4	Sepia Tone Example	19
3.1.1	Histogram Averaging Rate	9	5.3.5	User Palette Save - Load File Format	20
3.1.2	Histogram Maximum Percent Bin	9	5.4	Lens Distortion Correction	20
3.1.3	Gamma Correction.....	10	5.4.1	Tuning Lens Distortion Parameters.....	21
3.2	Color Enhancement.....	10	6	Questions and Additional Support.....	21
3.3	Sharpen Stage	10			

 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.



1 Overview

This document describes an overview of how to use the various enhancement techniques to improve the contrast and defined edges in streaming video.

1.1 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control (IDD) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

1.2 Sightline Software Requirements

IMPORTANT: The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.

1.3 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

Table 2: Software Feature Version Requirements

Software Function	Initial Software Release	Notes
Enhancement: CLAHE, LAP, Denoise, etc.	2.22.xx	CLAHE requires the Enhancement. CLAHE9 and CLAHE10 require both Enhancement and High Bit Depth/Temp Telemetry.
Enhancement Advanced	3.1.x	
False Color	2.22.xx	False color is included in Sightline software. Optional/recommended: High Bit Depth/Temp Telemetry



2 Rendering Video

2.1 Overview

Rendering is the process of generating output video from one or more camera sources and generating output video suitable for encoding. The analysis process is composed of registration, tracking, detection, etc. The rendering process is initiated after the analysis process is complete. Each stage of the rendering process may be disabled by the appropriate selection of parameters.

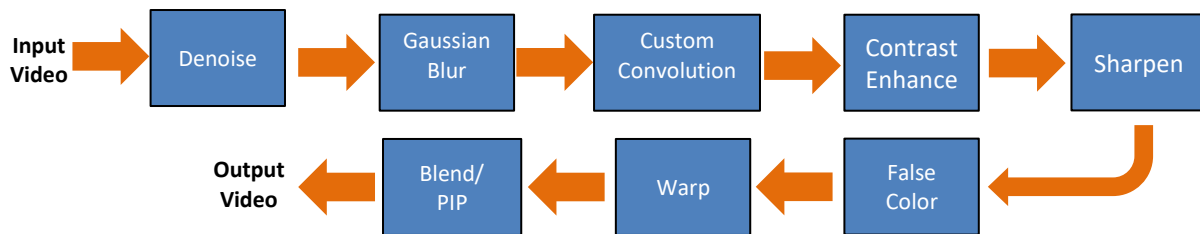


Figure 1: Rendering Video Process

Each frame of analyzed video is passed through the rendering process before encoding to network video output, digital display, or analog display:

- Gaussian blur applies a small spatial blur to the image. This is not commonly used.
- Denoising reduces video noise by computing a weighted average of the current frame of video with previous frames.
- Contrast enhancement applies to one of several different contrast enhancement algorithms to optimize the overall contrast of the image, e.g., CLAHE and LAP.
- Sharpen enhances edges and small details in the video.

The screenshot shows the 'Enhance' tab in the software interface. It contains various sliders and checkboxes for video enhancement. The 'Contrast Mode' is set to 'None'. The 'Strength' slider is at 18. The 'Sharpen' slider is at 0. The 'Sharpen Radius' is set to 1. The 'Blend' slider is at 255. The 'Denoise' slider is at 0. The 'Motion Mask' checkbox is unchecked. The 'Color Enhance' slider is at 0. The 'Scintillation Mode' is set to 'Manual'. There are checkboxes for 'Feature Based Histogram' and 'Square Root Histogram', both of which are unchecked. The 'Hist Ave Rate' slider is at 0. The 'Hist Max Pct' slider is at 0.0%. The 'Brightness' slider is at 128. The 'Contrast' slider is at 128. The 'Gaussian' slider is at 0. The 'LAPMinDiff' slider is at 0. There is a 'Reset All' button. Below the sliders, there is a section for 'Enhancement Saves' with a dropdown menu and a text input field for 'Save current enhance setting as'. At the bottom, there is a 'False Color' dropdown menu set to 'Default' and a 'Deconvolution Sigma' input field set to 0.

Figure 2: Enhance Tab - Panel Plus

The size of the features sharpen enhances can be changed with the Sharpen Radius value. The default value of 1 in maintains the standard sharpening effect. Increasing the value to 3 enhances slightly larger feature.

- Warp scales, rotates, and shifts video for stabilization and user-commanded effects.
- Blend/PIP combines processed video streams taken from two different sources in a variety of ways.



2.2 Denoise

Denoising is accomplished by registered frame averaging. In a system with a moving camera, it averages each frame of video after adjusting for frame offset motions, preventing the image from blurring out when the camera moves. This algorithm is most useful for uncooled IR sensors and in low visible light situations. The *Denoise* slider control in Panel Plus specifies the amount of averaging that is applied each frame:

- 0 = no denoising
- 64 = add 25% from the new frame / 75% from previous frames
- 128 = add 50% from each new frame / 50% from previous frames
- Suggested maximum value: ~240

Denoising is especially helpful when using contrast enhancement or sharpen. The algorithms in these stages tend to amplify image noise that denoising can suppress as shown in [Figure 5](#).

As a side effect of denoising, motion of foreground objects in the scene that are not captured by frame-to-frame registration will become blurry.

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *Denoising requires the Enhancement software feature. The Stabilization software feature is needed for registered denoising (for moving cameras).*

 **IMPORTANT:** In cameras with full 16-bit pixel depth, sparkle-pixel wrap artifacts can appear when running denoise and contrast enhancement. Please contact [Support](#) to discuss your application and potential solutions.

2.2.1 Motion Mask

Motion masking is used to mask moving objects from the denoise enhancement. Applying motion masking works by first detecting regions of motion in the scene. Portions of the scene with motion are then excluded from the averaging process used by denoise. Motion masking removes image noise and scintillation effects without blurring moving objects. Motion masking is also helpful in conjunction with CLAHE, LAP and edge sharpening. These algorithms amplify image noise that motion masking can suppress without blurring moving objects.

Set this function with the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *Motion mask requires the enhancement and enhancement advanced software feature. The stabilization software feature is needed for registered denoising (for moving cameras).*

The following parameters are related to and useful for motion mask, and can be found in the *Track* tab by clicking the *Advanced* button in the *Detect* tab.

The *NframesBack* setting determines how far back in time to search. A larger value allows the system to mask slower moving objects but slightly increases processing time.

Downsampling: More downsampling (4X or 8X) runs faster but may miss small objects that would get masked with 2X.

Average Time Constant determines the approximate amount of time to fully replace the background model. A larger time constant produces a model that is less affected by slow moving objects but will take longer to initialize when the camera is moved (for a PTZ).

Downsampling: More downsampling (4X or 8X) runs faster but may miss small objects that would get masked with 2X.



Figure 3: CLAHE and Sharpen Example



Figure 4: CLAHE and Denoising Example



Figure 5: Masked Denoising Example

2.3 Custom Convolution

New in 3.10.x software.

Custom convolution enhancement is intended for customers who want to apply their own convolution-based image enhancements, such as custom sharpening kernels or edge highlighting.

When a new kernel size is selected, the kernel defaults to a centered value of 1 with surrounding zeros. This configuration has no effect on the image.

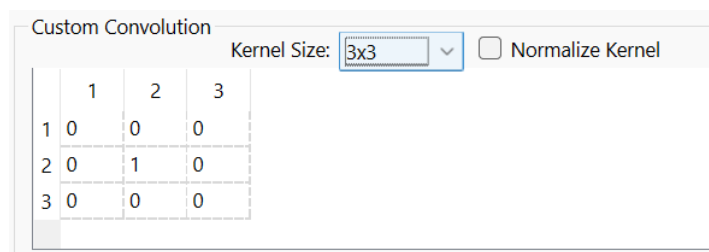


Figure 6: Custom Convolution



2.3.1 Custom Convolution Examples

Box Blur:

To average the surrounding pixels, use the kernel shown in **Figure 7** with *Normalize Kernel* selected.

Using the *Gaussian slider* in the *Enhancement* tab may be a better alternative.

Custom Convolution

Kernel Size: 3x3 ☒ Normalize Kernel

	1	2	3
1	1	1	1
2	1	1	1
3	1	1	1

Figure 7: Box Blur Example

Edge Highlighting:

The following kernel highlights the edges in the image. To display the highlighted edges on top of the original image, replace the center value of 40 with 41.

Custom Convolution

Kernel Size: 3x3 ☐ Normalize Kernel

	1	2	3
1	-10	0	-10
2	0	40	0
3	-10	0	-10

Figure 8: Edge Highlighting Example

2.3.2 Custom Convolution Performance Expectations

The performance of the custom convolution depends heavily on the kernel size and platform selection. The expected impact on timing is shown in **Table 3**. Note that the frame rates assume no other enhancements are applied.

Table 3: Custom Convolution Performance

4100-OEM				1750-OEM			
Image Size	Kernel Size	Convolution Timing (us)	FPS	Image Size	Kernel Size	Convolution Timing (us)	FPS
640x480	None		30	640x480	None		
	3x3	250	30		3x3	1400	30
	5x5	800	30		5x5	3950	30
	7x7	1500	30		7x7	6900	30
	9x9	2200	30		9x9	NA	NA
1920x1080	None		30	1920x1080	None		
	3x3	1500	30		3x3	9000	30
	5x5	4600	30		5x5	25500	28
	7x7	9000	30		7x7	45100	18
	9x9	10900	30		9x9	NA	NA

2.4 Contrast Enhancement

The contrast enhancement consists of one of the following basic algorithms: CLAHE, LAP, histogram equalization, or gamma correction. Select the desired mode from the *Contrast Mode* dropdown, or choose *None* to disable contrast enhancement



All contrast modes share the following two parameters:

- **Strength:** Has an effect that differs depending on the specific algorithm selected. See below.
- **Blend:** Allows mixing (alpha blending) of the output of the enhancement algorithm with the input to the Enhance Stage. This allows a user to soften the effects of the contrast enhancement stage.

0 = Show 100% of the original image.

128 = Show 50% of the original image combined with 50% of the enhanced image.

255 = Show 100% of the enhanced image.

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.



Figure 9: CLAHE, LAP, and Histogram Equalization Examples

2.4.1 CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) brings out hard to see (low contrast) features in video including video that contains brighter and darker areas as shown in [Figure 9](#).

Adaptive Histogram Equalization (AHE) is a computer [image processing](#) technique used to improve [contrast](#) in images. It differs from ordinary [histogram equalization](#) in the respect that the adaptive method computes several [histograms](#), each corresponding to a distinct section of the image, and uses them to redistribute the lightness values of the image. It is therefore suitable for improving the local contrast and enhancing the definitions of edges in each region of an image.¹

¹Wikipedia contributors, "[Adaptive histogram equalization](#)," *Wikipedia, The Free Encyclopedia*, (accessed May 7, 2020).



Contrast Limited AHE: Ordinary AHE tends to overamplify the contrast in near-constant regions of the image since the histogram in such regions is highly concentrated. As a result, AHE may cause noise to be amplified in near-constant regions. Contrast Limited AHE (CLAHE) is a variant of adaptive histogram equalization in which the contrast amplification is limited, to reduce this problem of noise amplification.¹

CLAHE: Standard CLAHE works with an 8-bit image input. For a 16-bit camera this will be the 8-bit image after autogain is applied.

CLAHE9 and **CLAHE10:** These versions use either 9 or 10-bits of a 16-bit input image as the input for the CLAHE operation. This can allow for improved contrast in the result image at the expense of slightly more processing time. CLAHE9 and CLAHE10 can also be applied to an 8-bit image input that has been denoised into a >8 bit per pixel intermediate image.

The *Strength* parameter controls the contrast limit of CLAHE. A larger value allows the algorithm to map a larger contrast but can also tend to bring out more noise.

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *CLAHE requires the Enhancement software feature. CLAHE9 and CLAHE10 require both the Enhancement and High Bit Depth/Temp Telemetry software features.*

2.4.2 Local Area Processing

Local Area Processing (LAP) brings out hard to see (low contrast) features shown in [Figure 9](#). LAP emphasizes differences from the local image average to provide an increased amount of detail in low contrast areas of a video or image. The algorithm makes details visible in underexposed or overexposed portions of the image.

The *Blend* parameter is usually used to vary how strong the LAP processing effect is. Generally, set *Blend* to a value of 128 or less.

The *Strength* parameter controls the size of the kernel used in the processing and a larger value provides more enhancement. Values are between 0 (minimal enhancement) and 18 (maximum enhancement). For LAP mode there is no additional enhancement benefits after 18.

The *LAPMinDiff* parameter is used to remove artifacts that may look like contour lines introduced by LAP in very flat scene areas. Setting a small value prevents LAP processing in these very low contrast areas.

These can all be set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

New in 3.10.x software.

LAP now supports the full bit depth of a 16-bit image. To enable this in Panel Plus, select the *Enhance* tab and then select *LAP 16* in the *Contrast Mode* drop down menu. The original LAP contrast mode uses only 8 bits of pixel data

 *LAP requires the Enhancement software feature.*

2.4.3 Histogram Equalization

Histogram equalization brings out hard to see (low contrast) features. It is like CLAHE, but it operates on the entire image rather than on subregions. It may not bring out some image features that CLAHE does if there are large shadows or uneven lighting.



This method usually increases the global **contrast** of many images, especially when the usable **data** of the image is represented by close contrast values. Through this adjustment, the **intensities** can be better distributed on the histogram. This allows for areas of lower local contrast to gain a higher contrast. Histogram equalization accomplishes this by effectively spreading out the most frequent intensity values.²

The *Strength* parameter controls the contrast limit of histogram equalization. A larger value allows the algorithm to map a larger contrast but can also tend to bring out more noise.

The *Brightness* parameter shifts the resulting histogram to be either brighter or darker.

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *Histogram equalization requires the Enhancement software feature.*

2.4.4 Feature Based Histogram Equalization

Feature-based histogram equalization emphasizes image areas with features and edges over flat backgrounds or uniform regions. In scenes with a small number of foreground objects on a flat background, background pixels can dominate the histogram in standard equalization. The resulting map may enhance noise in the background, which can obscure image features.

In this method, the histogram for pixel values that have a strong response to a high-pass filter are weighted more heavily, helping to emphasize image features over background areas

 *Feature based histogram equalization works with CLAHE/9/10 and histogram equalization. It does not work with LAP.*

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

2.4.5 Histogram Equalization + Gamma correction

This is the same as the histogram equalization shown above, but simultaneously applies a gamma correction so the resulting image histogram has a gamma-shaped curve.

This can be set using the [SetVideoEnhancementParameters \(0x21\)](#) command. The *enhanceParam* value controls the gamma value, and the *lapMinDiff* value sets the histogram equalization strength value.

 *See Gamma Correction for the range of values and behavior of the enhanceParam (gamma value).*

3 Square Root Histogram Equalization

This method can prevent large constant background or foreground areas in the image from dominating the histogram map. It applies a square root to the image histogram bins before applying histogram equalization.

 *Square root histogram equalization works with CLAHE/9/10 and histogram equalization. It does not work with LAP.*

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

² Wikipedia contributors, "[Histogram equalization](#)," *Wikipedia, The Free Encyclopedia*, (accessed May 7, 2020).



In **Figure 10**, background features are not visible in the original image. Histogram equalization makes the background visible, but the person appears saturated. Both feature-based and square root histogram equalization improve this by bringing the hot areas in the image out of saturation.

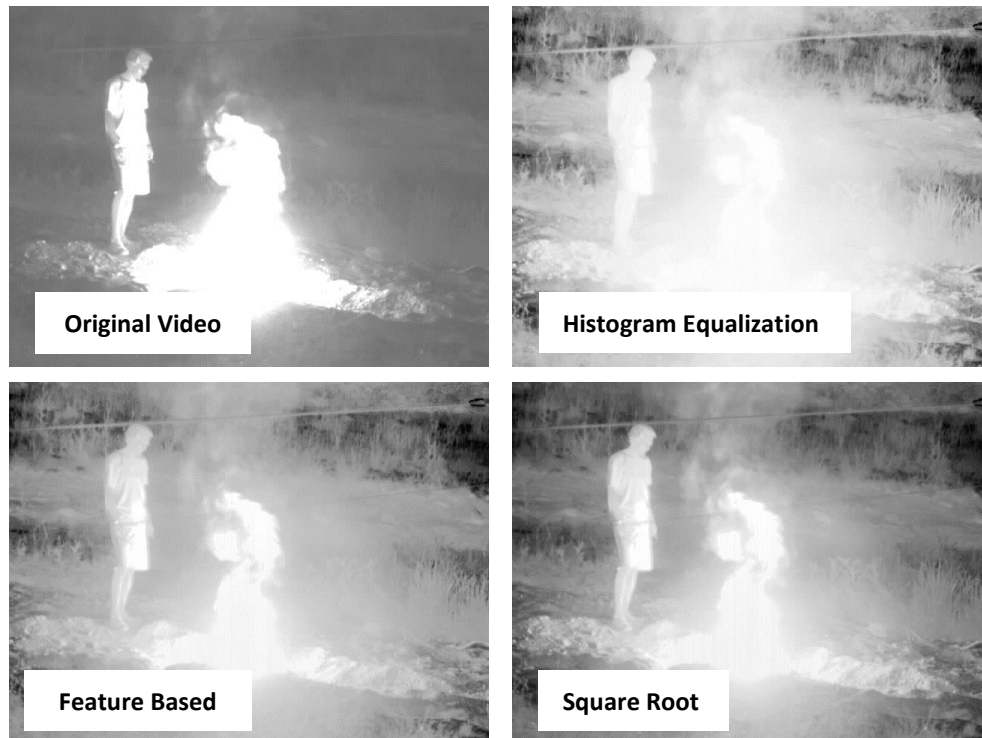


Figure 10: Histogram Equalization with Feature Based and Square Root Modifications Examples

3.1.1 Histogram Averaging Rate

Prevents the histogram equalization map from changing suddenly due to rapid scene brightness changes. The histogram equalization map is averaged by *histAveRate*/256 every frame. Setting the *Hist Ave Rate* to 0 fully replaces the histogram equalization map every frame. Setting *Hist Ave Rate* to 128 updates the histogram equalization map by 50% every frame.

 *Histogram averaging rate works with CLAHE and histogram equalization. It does not work with LAP.*

Set this function with the [SetVideoEnhancementParameters \(0x21\)](#) command.

3.1.2 Histogram Maximum Percent Bin

Limits the amount of histogram in a single bin to this percent. This prevents a single background or foreground gray-level from dominating the histogram map which can prevent other areas of the image from seeing contrast enhancement.

 *Histogram maximum percent bin works with CLAHE and histogram equalization. It does not work with LAP.*

Set using the [SetVideoEnhancementParameters \(0x21\)](#) command.



3.1.3 Gamma Correction

Performs a power law mapping which widens gray-level spacing in dark areas to better match human perception, especially of features in dark areas of a scene.

The *Strength* value controls the mapping where $\text{gamma} = \text{Strength} / 10$. A value of 1 corresponds to no change. Values < 10 will make the image brighter and values > 10 will make the image darker. The maximum value is 28.

3.2 Color Enhancement

Color enhancement helps the colors in a scene to stand out as shown in **Figure 11**. The *colorEnhance* parameter has a range of 0 (no color enhancement) to 255 (maximum color enhancement).

Typical values for outdoor scenes are between 100 and 200. Color enhancement is useful in combination with CLAHE or histogram equalization. It is also beneficial when the intensity of the scene has increased the color looks flat.

Color enhancement can be set using the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *Using color enhancement in brightly lit indoor scenes can make colors look overly saturated.*

 *Color enhancement requires the Enhancement software feature.*



Figure 11: Color Enhancement Example

3.3 Sharpen Stage

Edge sharpening is an image filtering technique that brings out edges in an image. The *Sharpen* parameter has a range of 0 (no sharpening) to 15 (maximum sharpening).

Set this function with the [SetVideoEnhancementParameters \(0x21\)](#) command.

Software version 3.9.x and above now includes a *Sharpen Radius* setting. It defaults to 1 to enhance very small details but can be changed to 2 or 3 to sharpen slightly larger features.

 *Edge Sharpening requires the Enhancement software feature.*

3.4 Region of Interest (ROI) processing

A region of interest may be specified that restricts enhancement processing to a sub-region of the video stream. Select a region in the camera input coordinates. The processing operations described above will apply only to that subregion. Use this mode to improve runtime performance or for special effects, such as to demonstrate the effect of enhancement features.



To specify the entire video region, set all four ROI parameters (Upper Left X/Row, Upper Left Y/Col, Height, Width) to 0.

Set this function with the [SetVideoEnhancementParameters \(0x21\)](#) command.

 *For the 1500-OEM and 3000-OEM, this setting is ignored when using sharpen/denoise without CLAHE, LAP or histogram equalization.*

 *In the 3.7.x software, enhancements will not be applied to a Picture-in-Picture (PiP) window of a camera using Region of Interest (ROI) when the main camera has the same index.*

3.5 False Color Stage

To use the *User Palette* mode, see the [False Color Palettes](#) section to setup a custom color user palette.

Set False Color parameters using the [SetDisplayParameters \(0x16\)](#) command.

 *Available in all software feature configurations.*

3.6 Multi-Camera Image Blend Stage

3000-OEM, 4000-OEM, and 4100-OEM only.

Blending is described in detail in [EAN-Blending](#).

 *Blending requires the Blending software feature.*

4 Application Notes

Some video enhancement settings can cause issues when streaming video. See the Troubleshooting section in [EAN-Encoding](#) for more information.

4.1 Warp Stage

Rotation is useful for gimbal systems where the image may be shown on its side or upside down.

The *Pan/Tilt* controls in the *Manipulate* section of the *Video* tab may be used to bore-sight an image with the lens, or to move the shown area of a larger image in a smaller output frame.

In some cases, a small zoom may be applied to clip off the edges of the image when there are stabilization artifacts.

Set using the [SetDisplayParameters \(0x16\)](#) command.

 *Available in all software feature configurations.*

 **IMPORTANT:** Do not use *Pan/Tilt* or *Roll* while blending the same camera. Use [SetMultipleAlignment \(0x74\)](#) to move and rotate the cameras.

4.2 Mitigating Scintillation Effects

When operating with high magnification imaging systems atmospheric effects can lead to distracting or disruptive visual effects such as heat shimmer, scintillation, and turbulence.

The *Enhance* tab contains settings that can help reduce scintillation and increase image clarity. *Denoise* primarily reduces scintillation, while the other parameters smooth and clarify the image.



4.2.1 Scintillation Reducing Parameters

- **Denoise:** This averages the image over time with image-to-image registration to dampen the scintillation effect. Normal use values range from around 100-245. Reduce this number to decrease the dampening effect. Small changes to this parameter can have a significant impact.
- **Motion Mask:** Prevents denoise from blurring out moving objects in the scene. If there are no moving targets in the scene uncheck this option.
In 3.5.x software this feature is labeled Denoise Mask.
- **Contrast Mode (CLAHE or LAP):** Enhances contrast within the image to improve clarity. Also try CLAHE9 or CLAHE10 for a high bit depth camera.
- **Sharpen:** Use a low value to enhance edges. Higher values may cause grainy artifacts.
- **Blend:** When the image appears over-enhanced, a lower setting will reduce the amount of enhancement. This is required for LAP mode, which has a recommended starting point of 80 to 150. A CLAHE setting of 255 is a good starting point, but the video may look better if this value is reduced.
- **Hist Max Pct:** Used to mitigate hyper-saturation. If one color is overly dominant in the image increasing this value can help alleviate that issue.

Scintillation Mitigation - 3.6.x software and above:

1. From the *Enhance* tab go to the *Scintillation Mode* dropdown menu.
2. Make a selection from the drop-down list that most accurately represents the type of video (IR = Infrared).

For IR video the non-IR alternatives can occasionally perform better.

3. If there are no moving targets in the scene uncheck the *Motion Mask* option.
4. Set the *Scintillation Mode* to *Manual* and adjust the [scintillation reducing parameters](#) to optimize clarity and minimize distortion.

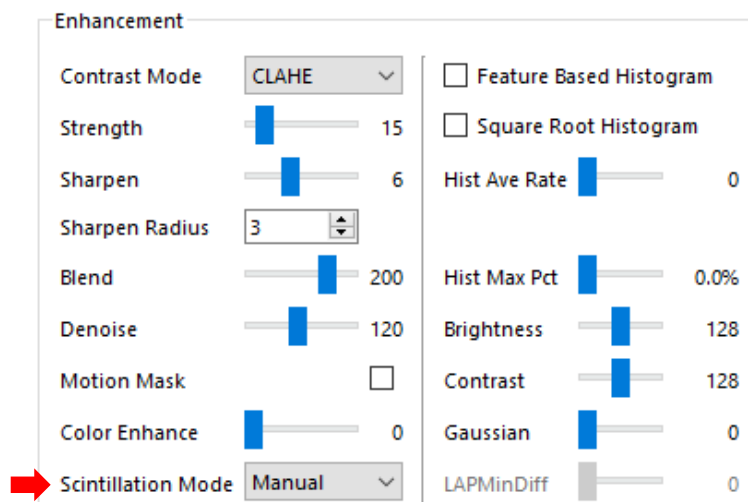


Figure 12: Scintillation Mitigation Controls - 3.6.x Software and Above



Scintillation Mitigation - 3.5.x software:

1. In the *Scintillation Mitigation* drop-down menu use one of the pre-fill parameters for reducing scintillation effects.
2. After choosing an option, adjust the [scintillation reducing parameters](#) to optimize clarity and minimize distortion.

Denoise Mask prevents *Denoise* from blurring out moving objects in the scene. This works best with *Denoise Mask* » *Aerial*.

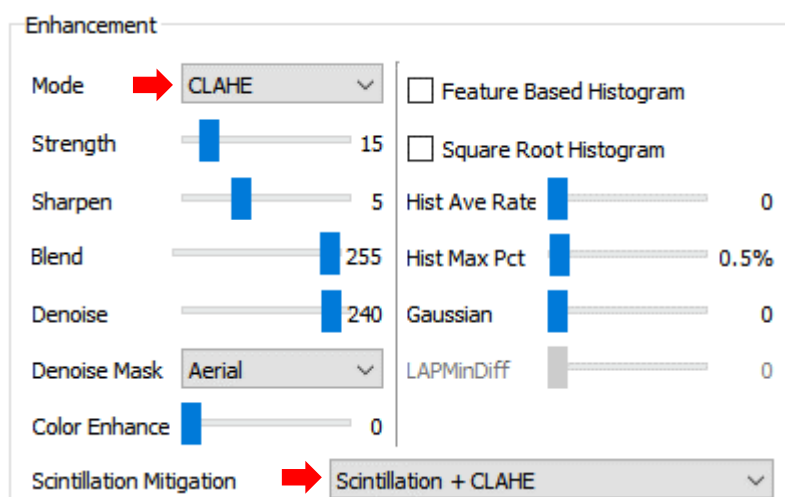


Figure 13: Scintillation Mitigation Controls - 3.5.x Software

4.3 Saving Enhancements

New in 3.6.x software and above.

In addition to saving the enhancement settings to the board, enhancements can also be saved to macros on the PC. This can be helpful if the camera often points at similar scenes, or switch between two different enhancement settings quickly.

The *Save current enhance setting as* button can be used to name and record the current enhancement settings for later use. The recordings can be found in the *Enhancement Saves* dropdown menu.

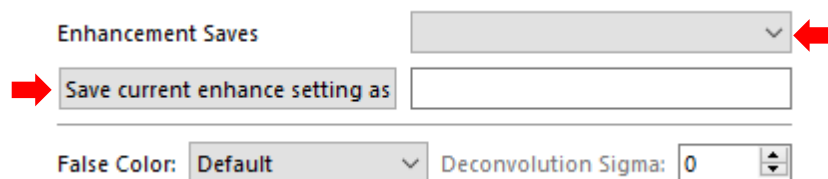


Figure 14: Save Current Enhancements

Ensure that the *Macro Destination Macro* setting matches the connection port that is being used, or *Enhancement Saves* will not function. See the *Panel Plus User Guide* for more information.



4.4 Replicating Effects of IR Cameras

The following table displays the FLIR camera features and their corresponding Sightline parameters for the high bit depth autogain min and max calculations. These also include histogram equalization and other enhancement controls.

Table 4: FLIR Camera and Sightline Parameters for Autogain Calculation

FLIR Camera Features	Sightline High Bit Depth Autogain and Enhancement Parameters
Information-Based Equalization	Feature based histogram equalization. (Create histogram from high-pass energy)
Adaptive Contrast Enhancement (ACE)	Similar effect to using gamma correction enhancement mode.
Digital Detail Enhancement (DDE)	Implement via sharpen, LAP, CLAHE.
Linear percent	Similar effect to blend % when used with histogram equalization or CLAHE enabled.
Tail rejection	The high bit depth autogain dark and light tail rejection.
Damping factor	The high bit depth autogain auto rate and histogram averaging rate with histogram equalization or CLAHE.
Max gain	The high bit depth autogain contrast parameter.
Plateau values	Similar effect to <i>hist max percent</i> parameter when used with histogram equalization or CLAHE.

5 False Color

Using false color during video processing can help visually interpret and analyze video data in different ways. The best example is with a thermal imager, coloring hot objects in red and cold objects in blue for an intuitive presentation. Applying false color to grayscale video, such as from a thermal camera, when using the Blending feature can help users differentiate between the source of the data they see in the blended image.

5.1 False Color Guidelines

The value of false color added to thermal imagery can be subjective in certain scene situations. It is helpful to have two to three options that work best in most user case thermal imaging scenarios as shown in [Figure 15](#).

Find color patterns that allow objects of interest to be easily discerned using color.

Requires some experimentation with different colors and imagery.

False color can also be used in Electro-Optical (EO) imaging for detecting over exposure (saturation). For example, using the user defined color palette, a pattern can be defined that turns red with pixel values at 90% of the full saturation ($255 * 0.90 = 229$) allowing some flexibility to handle events that might be too bright or too dark.



Figure 15: False Color Image Example

5.1.1 Frequently Used Modes

White Hot / Black hot:	Imagery produced by a thermal camera where white is hot and black is cold or white is cold and black is hot.
Iron256:	Black is cold, violet and blue are warmer, progressing to reds and yellows, white being the hottest.
Ice / Fire:	Used to detect saturation points at either end of the intensity scale. 0 = blue = very cold things, 255 = red = hot things.
Jet/HotHSV/470CLR/Color1:	Provide a continuous spectrum of color.

5.2 Related Sightline Commands

SetDisplayParameters (0x16)	Allows user to choose 1 of many predefined false color palettes.
Set User Palette (0x72)	Allow users to define their own Y, U, V values for a custom false color palette.
SetDetectionParameters (0x2D)	Histogram detection (temperature) for moving targets.
Tracking Box Pixel Stats (0x78)	Periodic report of pixel statistics within track boxes.
Set Video Enhancement Parameters (0x21)	Different techniques to improve the contrast and defined edges in the video.



5.3 False Color Palettes

The *False Color* palette pull down menu can be accessed from the *Enhance* tab in Panel Plus. From this menu various color palettes can be selected. An example of each color palette is shown below.

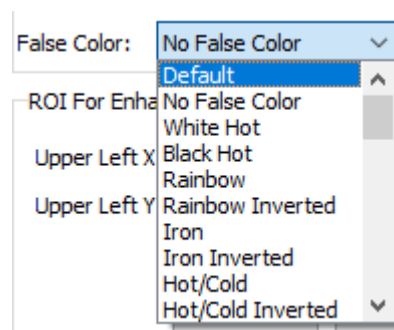


Figure 16: False Color Palette Menu

Table 5: False Color Palette

Palette Description	Palette Color Range
White Hot	
Black Hot	
Rainbow	
IRON	
Hot/Cold	
Jet	
Hot & HSV	
470CLR_S & Color 2	
Color 1	
Color 3 & IDDef	
HotIron	
IceFire	
Iron256	
Rain256	
Xvolcano	
Red	
Green	
Blue	



5.3.1 Custom Color User Palette

To create a custom false color palette:

1. From the main menu in Panel Plus go to *Configure » User Palette*.
2. Select a color. (Blue is used in this example.)

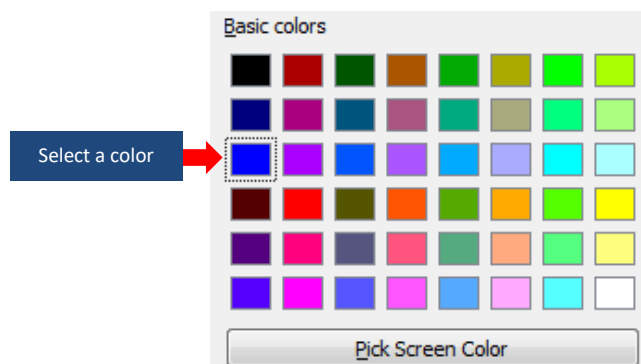
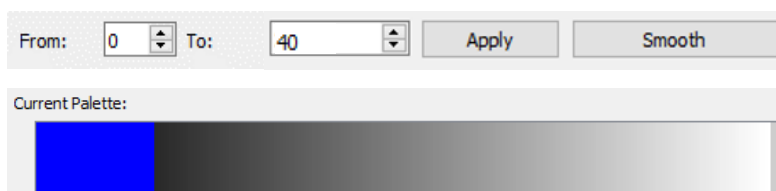


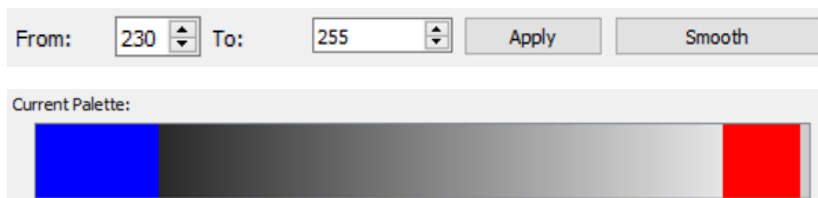
Figure 17: Custom Color User Palette

3. Set a range of grayscale values. (For this example, 0 to 40 is used). Click *Apply*. The updated color scheme is shown in the *Current Palette* window.



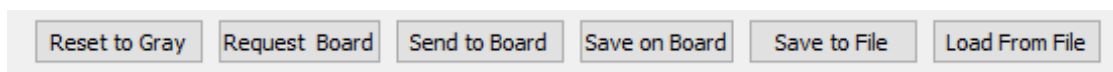
New colors can be set anywhere along the grayscale using the grayscale values from 0 to 255 (max). 0 starts the new color at the darker side of the grayscale. In this example the blue color has a gray scale value of 0 to 40.

4. Select another color (red is used in this example). Set the grayscale value to 230 to 255. This puts the color at the lighter end on the grayscale color range. Click *Apply*.



New in 3.6.x software: To smooth the transitions between the bands of color between From and To, click the Smooth button multiple times.

5. If no other adjustments are needed, click *Send to Board* to upload the new user palette. To start over click *Reset to Gray*.





- To view the custom user palette applied to the video image, select *User Palette* in the *False Color* menu.

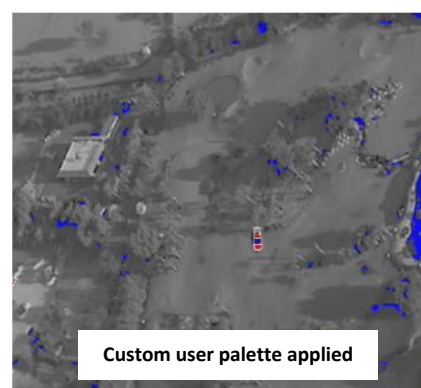
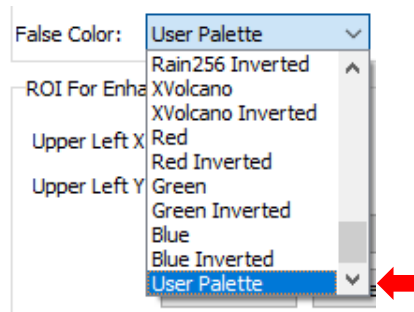


Figure 18: Custom User Palette

5.3.2 Custom User Palette with Pre-existing LUT

There is also the option to start with one of the existing user palettes in the *Start with This Palettes* drop-down menu. In the example shown in [Figure 19](#) *Iron Inverted* is selected. These palettes can also be edited as needed and then sent to the OEM as a custom palette. The default palettes in the menu will not change.

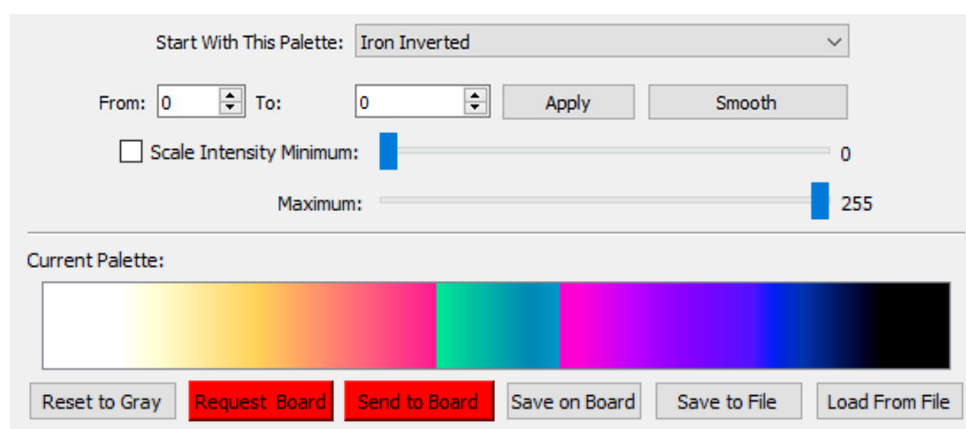


Figure 19: Existing User



5.3.3 Multiple Color Range Example

Multiple color ranges can also be defined in the user palette as shown in **Figure 20**.

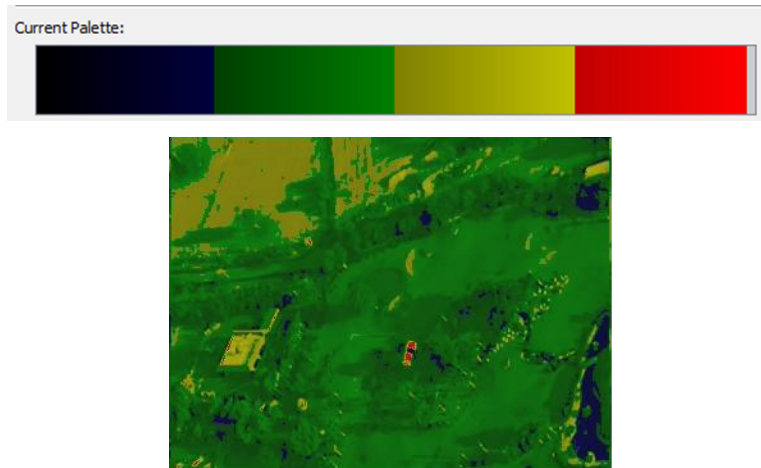


Figure 20: Custom Multiple Color User Palette

5.3.4 Sepia Tone Example

To apply a Sepia tone to an image:

1. Set the *Red*, *Green* and *Blue* values as shown.
2. Set the gray scale value to 0 to 255.

To automatically generate shades of the selected color, use the *Scale Intensity* option.

3. Click *Apply* and then *Send to Board*.

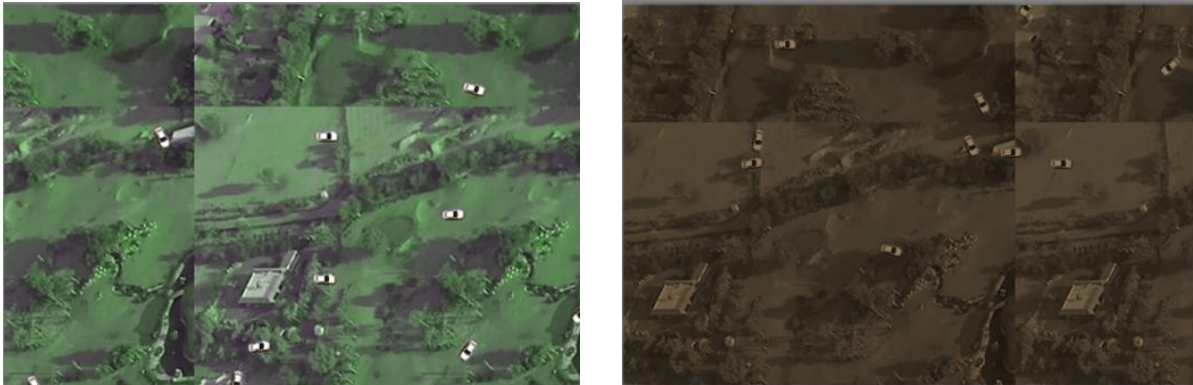


Figure 21: Sepia Tone Custom User Palette Applied

5.3.5 User Palette Save - Load File Format

The user palette can be saved to and loaded from a file using the *Save to File* and *Load From File* buttons in the *Edit User Palette* dialog window.

The file is binary and consists of 256 x 3 YUV 8-bit values. These values are the YUV color space mapping from gray levels 0 to 255 to intensity and color values.

Binary data in the user palette file is in the following order (all values are 8-bit unsigned characters):

Y[0] U[0] V[0] Y[1] U[1] V[1] ... Y[255] U[255] V[255]

 Many programs, such as ImageJ, store RGB values in their LUT files. Corresponding YUV values can be calculated from these RGB values using the YUV color model.³

$$Y = 0.299 * R + 0.587 * G + 0.114 * B;$$

$$U = -0.14713 * R - 0.28886 * G + 0.436 * B + 128.0;$$

$$V = 0.615 * R - 0.51499 * G - 0.10001 * B + 128.0;$$

5.4 Lens Distortion Correction

Certain lenses, particularly wide-angle lenses, introduce barrel distortion. During stabilization, distortion varies across the displayed frame as the sampled region shifts within the source frame. To avoid this variation, apply distortion correction before stabilization.

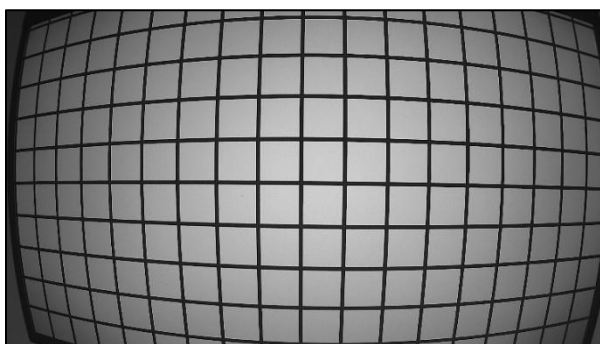


Figure 22: Barrel Distortion

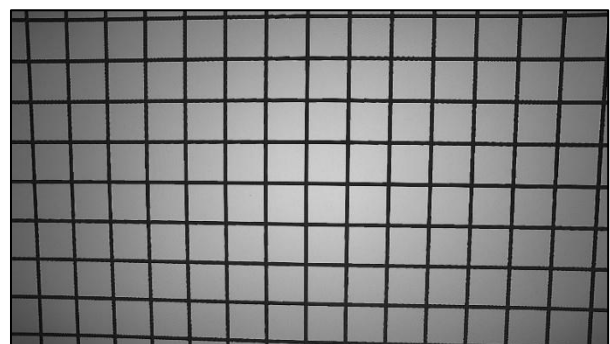


Figure 23: Corrected Barrel Distortion

³ Wikipedia contributors, "YUV," Wikipedia, The Free Encyclopedia, (accessed January 31, 2023).



5.4.1 Tuning Lens Distortion Parameters

To roughly tune the distortion amount parameter, find a straight edge in the scene and position it near the edge of the image. A good starting value is 0.05. If the edge still curves toward the center of the frame, increase the value. If the edge curves away from the center, decrease the value.

To fine-tune the distortion parameters, point the camera at an image of a grid. Start by adjusting the distortion amount until the lines near the edges appear straight. If one edge curves inward while the opposite edge curves outward, this indicates that the distortion center is offset from the image center. Adjust the center row and center column parameters until the distortion is symmetrical, then refine the distortion amount until all edges appear straight.

6 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).